

Algebra 1
Unit 6
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

Alanna made a mistake when solving the inequality $\frac{-7n+15}{8} - 5 \geq 3$. For questions 1-4, use Alanna's work below.

Given Inequality	$\frac{-7n+15}{8} - 5 \geq 3$
Step 1	$-7n + 15 - 5 \geq 24$
Step 2	$-7n + 10 \geq 24$
Step 3	$-7n \geq 14$
Step 4	$n \leq -2$

1. Identify the step in which Alanna made an error. Write a note to Alanna explaining why that step was incorrect.

Alanna made an error in Step 1.

Alanna, when "un-doing" the division of 8, you must multiply every term in the inequality by 8. You forgot to multiply the (-5) by 8.

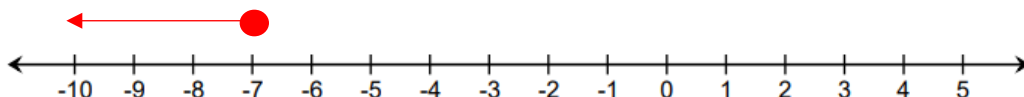
2. Solve the inequality $\frac{-7n+15}{8} - 5 \geq 3$.

$$\begin{array}{l}
 \frac{-7n+15}{8} - 5 \geq 3 \\
 \quad +5 \quad +5 \\
 \left(\frac{8}{1}\right) \frac{-7n+15}{8} \geq 8 \left(\frac{8}{1}\right) \\
 -7n + 15 \geq 64 \\
 \quad -15 \quad -15
 \end{array}
 \quad
 \begin{array}{l}
 \frac{-7n}{-7} \geq \frac{49}{-7} \\
 \boxed{n \leq -7}
 \end{array}$$

3. Write the solution in set-builder notation.

$$\{n \in \mathbb{R}: n \leq -7\}$$

4. Graph the solution set.



Use the inequality $0.06m + 2.56 > -0.15m + 8.86$ to complete questions 5-7.

5. Solve the inequality.

$$\begin{array}{r} 0.06m + 2.56 > -0.15m + 8.86 \\ +0.15m \qquad \quad +0.15m \end{array}$$

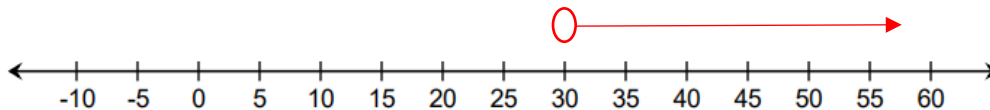
$$\begin{array}{r} 0.21m + 2.56 > 8.86 \\ -2.56 \quad -2.56 \end{array}$$

$$\frac{0.21m}{0.21} > \frac{6.3}{0.21} \qquad m > 30$$

6. Write the solution in set-builder notation.

$$\{m \in \mathbb{R}: m > 30\}$$

7. Graph the solution set.



Use the inequality $-\frac{1}{2}(5n + 2) \leq -\frac{1}{3}(10n - 2)$ to complete questions 8-10

8. Solve the inequality.

$$\left(\frac{6}{1}\right)\left(-\frac{1}{2}(5n + 2)\right) \leq \left(\frac{6}{1}\right)\left(-\frac{1}{3}(10n - 2)\right)$$

$$-3(5n + 2) \leq -2(10n - 2)$$

$$\begin{array}{r} -15n - 6 \leq -20n + 4 \\ +20n \qquad +20n \end{array}$$

$$5n - 6 \leq 4$$

$$\begin{array}{r} 5n - 6 \leq 4 \\ +6 \quad +6 \end{array}$$

$$5n \leq 10$$

$$n \leq 2$$

9. Write the solution in set-builder notation.

$$\{n \in \mathbb{R}: n \leq 2\}$$

10. Graph the solution set.

